Experimental Analysis of Public Policies

Session 5

Difference-in-Differences: 2x2 and TWFE

Etienne Dagorn
University of Lille

Session Objectives

By the end of this session, students should be able to:

- explain why before-after and post-treatment level comparisons can fail;
- compute a 2x2 DiD estimator from group means;
- label the 2x2 estimand as an ATT for the treated group in the post period;
- state parallel trends in plain language and potential-outcomes language;
- write and interpret the basic DiD regression;
- explain why serial correlation motivates clustered standard errors;
- use pre-trends, placebo outcomes, and anticipation checks as diagnostics;
- know when simple TWFE has a clear interpretation.

Roadmap

Policy Timing And Failed Comparisons

The 2x2 Double Difference

Regression And Simple TWFE

Diagnostics And Inference

Practice And Bridge

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From Instruments To Policy Timing

- Session 4 used an instrument Z to create exogenous variation in treatment received D.
- Some policy questions have no credible instrument at the individual level.
- They do have variation in when and where a policy begins.
- DiD asks whether untreated units reveal the missing trend for treated units.

Central question

Can untreated trends reveal the missing counterfactual trend?

Policy Example

- A region introduces a youth employment subsidy in 2025.
- Other similar regions do not introduce it.
- Youth employment rises in the treated region after 2025.
- Is the rise due to the subsidy or to a general recovery?

Counterfactual question

How would youth employment in the treated region have changed after 2025 without the subsidy?

Failed Comparison 1: Before And After

$$\bar{Y}_{T,1} - \bar{Y}_{T,0}$$

- This uses the treated group as its own comparison.
- It is tempting because it follows the same region over time.
- It fails if employment would have changed anyway because of recovery, recession, seasonality, or demographic change.

What is missing?

The treated region's untreated post-policy outcome, not just its pre-policy outcome.

Failed Comparison 2: Post-Treatment Levels

$$\bar{Y}_{T,1} - \bar{Y}_{C,1}$$

- This compares treated and untreated regions after the subsidy starts.
- It fails if treated and control regions had different baseline levels before the policy.
- A poorer treated region can still look worse after a successful policy.

What is missing?

The permanent treated-control difference that existed before treatment.

DiD Workshop: Build The Counterfactual

Scenario: a region introduces a youth employment subsidy in 2025.

1. Draw the treated-region employment path before and after 2025.
2. Draw the control-region path.
3. Use the control change to project the treated region's untreated post-2025 outcome.
4. State the exact trend assumption needed.

> **Debrief**
>
> The control group does not need the same level as the treated group. It must provide the untreated trend for the treated group.

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The 2x2 Setup

	Before	After
Treated group	$\bar{Y}_{T,0}$	$\bar{Y}_{T,1}$
Control group	$\bar{Y}_{C,0}$	$\bar{Y}_{C,1}$

- One treated group ($A_i = 1$), one control group ($A_i = 0$).
- One pre-treatment period, one post-treatment period.
- Treatment starts between the two periods.
- In this 2x2 setting, policy exposure is $D_{it} = A_i \times Post_t$.

Table Computation

	Before	After	Change
Treated	50	58	$58 - 50 = 8$
Control	55	59	$59 - 55 = 4$
DiD			$8 - 4 = 4$

Interpretation

Under the design assumptions, the population DiD identifies the post-period ATT; the sample contrast estimates it.

The DiD Estimator

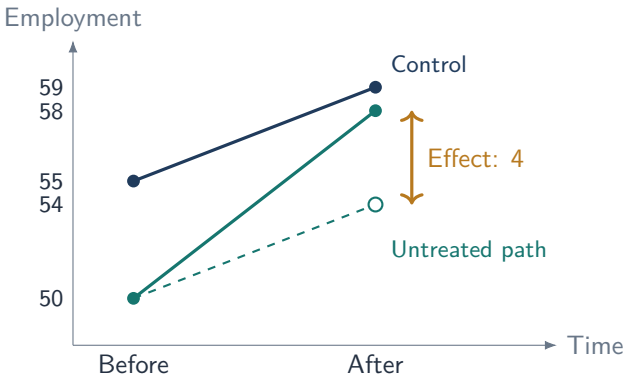
$$\widehat{\tau}_{DiD} = (\bar{Y}_{T,1} - \bar{Y}_{T,0}) - (\bar{Y}_{C,1} - \bar{Y}_{C,0})$$

- Treated change: what happened to the treated group.
- Control change: the proposed untreated counterfactual trend.
- Double difference: the extra treated change after removing the common trend.

Estimand

The usual 2x2 causal target is $ATT_1 = E[Y_{i1}(1) - Y_{i1}(0) \mid A_i = 1]$.

Draw the Missing Untreated Path



Counterfactual:

$$50 + (59 - 55) = 54.$$

Effect estimate:

$$58 - 54 = 4.$$

The dashed path is assumed; we never observe it for treated units.

Parallel Trends

Parallel trends

In the absence of treatment, the treated and control groups would have experienced the same average change in outcomes.

$$E[Y_{i1}(0) - Y_{i0}(0) \mid A_i = 1] = E[Y_{i1}(0) - Y_{i0}(0) \mid A_i = 0]$$

- Groups can have different levels.
- What matters is the untreated counterfactual trend.
- Pre-treatment trends are evidence about credibility, not the assumption itself.

Why the Treated Change Is Not Yet an Effect

Add and subtract the missing untreated post-policy mean:

$$E[Y_{i1} - Y_{i0} \mid A_i = 1] = \underbrace{E[Y_{i1}(1) - Y_{i1}(0) \mid A_i = 1]}_{ATT_1} + \underbrace{E[Y_{i1}(0) - Y_{i0}(0) \mid A_i = 1]}_{\text{change that would occur without the policy}} .$$

- Consistency links observed and potential outcomes.
- No anticipation makes the pre-period outcome untreated.
- The last term is missing: DiD substitutes the control group's change.

Parallel Trends Removes the Remaining Bias Term

Write $\Delta Y_i(0) = Y_{i1}(0) - Y_{i0}(0)$. With untreated controls:

$$DiD = ATT_1 + \underbrace{E[\Delta Y_i(0) \mid A_i = 1] - E[\Delta Y_i(0) \mid A_i = 0]}_{\text{bias from different untreated trends}}.$$

Identification

Parallel trends sets the bracketed difference to zero: $DiD = ATT_1$.
In a sample, $\widehat{\tau}_{DiD}$ estimates this population contrast.

Check: if the treated region would have gained 2 extra points anyway, how does a DiD estimate of 4 relate to its true ATT?

Three More Questions Before Calling It Causal

Condition	Subsidy example	Useful evidence
No anticipation	Firms do not react before the chosen start date	Announcement and implementation dates
No spillovers to controls	Jobs are not merely displaced from the control region	Commuting zones and neighbouring areas
Stable comparison	Migration or attrition does not change who is measured	Follow the same units; inspect composition

Design audit

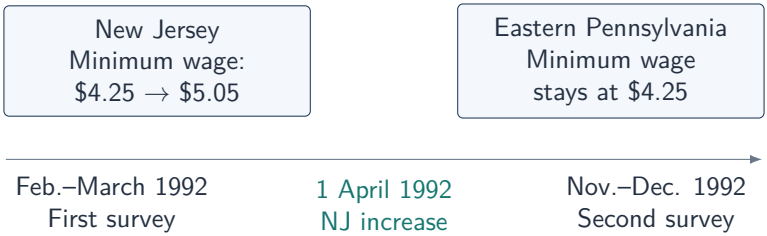
Choose the most plausible problem. Propose a check that could reveal it.

Levels Can Differ

- Treated regions may be poorer than control regions.
- Treated schools may have lower baseline scores.
- DiD can allow fixed level differences.
- It cannot allow different untreated trends unless those trends are explicitly modelled and justified.

> **Shortcut**
>
> DiD removes permanent group differences and common shocks, not group-specific shocks.

A Paper's Policy Question: Minimum Wages and Jobs



Question: did the increase reduce fast-food employment?
Design: survey 410 restaurants before and after; compare changes across states.

Card and Krueger (1994), AER, pp. 772–775.

Card and Krueger: Read the Result with Its Design

Outcome: full-time-equivalent (FTE) employment per restaurant.

Published contrast, Table 3	FTE employees per restaurant
NJ employment change minus PA employment change	$+2.76$
Reported standard error	1.36

- The estimate is positive; the study does not find the predicted job loss.
- A causal reading requires comparable untreated state trends and no anticipation.
- Two survey waves cannot establish pre-trends; measurement and state shocks matter.

Scope: these restaurants and this wage increase in 1992; not every labour market.

Card and Krueger (1994), Table 3, row 3, column (iii), and p. 778.

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The Regression Is the Same Four-Cell Comparison

$$Y_{it} = \alpha + \beta A_i + \lambda Post_t + \tau \underbrace{(A_i \times Post_t)}_{D_{it}} + u_{it}.$$

Fitted group mean	Before	After
Control	$\widehat{\alpha}$	$\widehat{\alpha} + \widehat{\lambda}$
Treated	$\widehat{\alpha} + \widehat{\beta}$	$\widehat{\alpha} + \widehat{\beta} + \widehat{\lambda} + \widehat{\tau}$

Key takeaway

$\widehat{\tau}$ equals the hand-calculated double difference.
The equation estimates the comparison; parallel trends gives it a causal interpretation.

Fixed Effects In The Simple Panel

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + u_{it}$$

- Unit fixed effects α_i absorb time-invariant differences across units.
- Time fixed effects λ_t absorb shocks common to all units in a period.
- Identification comes from within-unit changes relative to common calendar changes.
- Fixed effects do not absorb a shock that affects only treated units when treatment starts.

Adding Controls

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + X'_{it}\gamma + u_{it}$$

- Pre-treatment covariates can improve precision or define a conditional comparison.
- Clearly exogenous time-varying controls may support conditional parallel trends.
- Controls affected by treatment can change the estimand or bias the estimate.
- Controls do not replace the need to defend parallel trends.

When Does One TWFE Coefficient Answer the Question?

- In the balanced 2x2 panel, the coefficient equals the simple double difference.
- With one adoption date and several post-periods, explain which periods receive weight.
- A single average can hide a delayed gain or a temporary loss.
- Defend untreated trends, no anticipation and the comparison population separately.

Diagnostics are evidence, not a condition that guarantees validity

Insignificant pre-trend coefficients cannot establish parallel post-policy trends.

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Inference: Many Observations Can Mean Few Policy Shocks

Region A: one policy decision Many workers, repeated years	Region B: one policy decision Many workers, repeated years

- Outcomes within a region and over time can be correlated.
- Standard errors should respect the assignment and dependence structure.
- Few independent clusters make inference difficult, even with many workers.

Bertrand, Duflo and Mullainathan show severe over-rejection with conventional standard errors in placebo-law exercises.

Bertrand, Duflo and Mullainathan (2004), QJE; author manuscript.

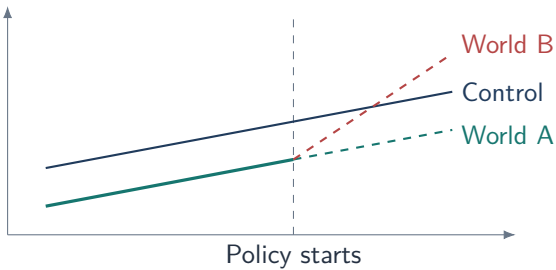
Pre-Trends

- Multiple pre-treatment periods allow a visual pre-trend check.
- Similar pre-trends support parallel trends but do not prove it.
- Diverging pre-trends weaken credibility.
- Very noisy pre-trends may be uninformative.

Exam phrase

The pre-trend evidence is supportive but cannot rule out a treated-specific shock after treatment.

Two Worlds Can Have the Same Pre-Trends



World A: untreated trends remain parallel.

World B: a new factory opens only in the treated region.

Both worlds fit the same pre-policy evidence.

Diagnostics do not establish identification

Flat pre-trends cannot rule out a treated-specific shock after implementation.

Placebo Checks

- Pretend treatment happened before it actually did.
- Test outcomes that should not respond to the policy.
- Use groups that should not be affected.
- Placebos are diagnostic evidence, not automatic validation.

Good placebo

Choose a false timing, group, or outcome that would reveal the most plausible violation of parallel trends.

Design-Threat Lab

Threat	Question to ask
Different trends	Why would untreated trends be parallel?
Anticipation	Did behaviour change before treatment?
Spillovers	Did controls get indirectly treated?
Composition	Did the groups change over time?

Student output

Pick the most credible threat in the subsidy example and propose one diagnostic or design response.

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Applied Reading Questions

- What is the treated group and treatment date?
- What comparison group is used?
- What ATT or treated-period target is being estimated?
- What is the evidence for parallel trends?
- What standard errors are reported?
- What threat remains most credible?

R Mini-Lab: Recover the Same Contrast Two Ways

Question: what is the subsidy effect in the simulated two-period panel?

```
source("r_exercises/week_05_did_twfe.R")
m <- with(panel, tapply(Y, list(treated, post), mean))
did <- (m["1","1"] - m["1","0"]) -
       (m["0","1"] - m["0","0"])
fit <- lm(Y ~ treated * post, data = panel)
c(by_hand = did, regression = coef(fit)["treated:post"])
```

Eight minutes: calculate, then qualify

Explain the equality. State the assumption needed for a causal interpretation.
The simulated Y is an outcome score: report **score units**, not percentage points.

Exercise Bridge

- Theory sheet: `homework/session_05_homework.md`.
- QCM: `assessments/qcm/session_05_qcm.md`.
- Applied exercise: compute and critique a 2x2 DiD.
- Exam skill: defend or reject parallel trends in a policy scenario.

> **Exercise rhythm**
> Compute first, interpret second, then diagnose the assumption.

Transition To Session 6

- Today one DiD coefficient summarised one before-after contrast.
- Many policies have effects that appear slowly, fade, or change as institutions adapt.
- Many policies are adopted by different units in different years.

Bridge question

What changes when treatment effects evolve and units adopt at different times?

Design Checklist

1. Who is treated and when?
2. Who is the comparison group?
3. What ATT or treated-period estimand is targeted?
4. What is the identifying trend assumption?
5. What diagnostics support it?
6. What inference level is appropriate?
7. What remaining shock could break the design?

Key Takeaways

- DiD compares changes, not levels.
- The clean 2x2 estimand is usually the ATT for treated units in the post period.
- Parallel trends is the core identifying assumption.
- Fixed effects remove permanent unit differences and common time shocks.
- Pre-trends and placebos help diagnose credibility but do not prove identification.
- Standard errors must respect the panel and treatment structure.

Appendix: Conditional Trends Require Conditional Comparisons

For baseline covariates X, assume equal untreated changes at each x:

$$E[\Delta Y_i(0) \mid A_i = 1, X_i = x] = E[\Delta Y_i(0) \mid A_i = 0, X_i = x].$$

Define $m_0(x) = E[\Delta Y_i \mid A_i = 0, X_i = x]$. Then:

$$ATT_1 = E[\Delta Y_i - m_0(X_i) \mid A_i = 1].$$

- Average the control trend over the **treated** covariate distribution.
- Need $\Pr(A_i = 0 \mid X_i = x) > 0$ on that treated support.
- A raw control mean or an arbitrary additive X term need not implement this.

References: Empirical Design and Inference

- Card and Krueger (1994). "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review*, 84(4), 772–793.
  Author-hosted paper.
- Bertrand, Duflo and Mullainathan (2004). "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics*, 119(1), 249–275.
  Author manuscript.